OSS CAPSTONE AUDIT — Git (Version Control System)

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Abstract

This project is about Git. Git is a tool that helps people work together on projects. The goal of this project is to learn about Git from a point of view and how it helps open-source projects. We will talk about where Git came from how it was made and how it affects the community of developers. Git helps developers track changes and work together. This project also includes five scripts that show how to use Linux commands in real-life situations.

These scripts demonstrate how to use Linux commands to do things like monitor the system check packages and analyze logs. They show how Linux commands can be used in real-world situations. The project emphasizes the importance of open-source software. It promotes working being transparent and coming up with new ideas. Through this project students learn about concepts. Get hands-on experience.

Introduction to Open Source Software

Open Source Software is software that anyone can use, study and modify for free. It gives users the freedom to access the code and make changes as needed. This makes it easier for students and developers to learn and develop.

Unlike paid software Open Source Software does not depend on one company. Many developers from around the world contribute to improve it. This teamwork helps fix bugs and add new features.

Open Source Software is widely used in education, research and companies. It is also trusted for its transparency and security. Overall Open Source Software promotes working being creative and coming up with new ideas in technology..

Introduction to Git

Git is a tool that helps developers manage and track changes in their code. It allows multiple people to work on the project without getting confused. One important feature of Git is that it is distributed, meaning everyone has a copy of the project. This makes it safe and reliable.

Git also supports branching, which lets developers work on features separately. After completing work changes can be merged easily. It is fast, efficient and

widely used in software development. Git helps teams stay organized and avoid mistakes. That's why it is a tool for developers.

Part A — Philosophy

A1: History and Origin of Git

Git was created by Linus Torvalds in 2005. At that time, Linux developers were using a tool called BitKeeper. When BitKeeper stopped being free, there was a need for a new system. Linus Torvalds developed Git to solve this problem. He designed it to be fast, simple, and open-source.

Git quickly became popular because of its performance and flexibility. It allowed developers to work independently and still collaborate easily. Over time, Git became the most widely used version control system. Today, it is used in almost every software project. Its creation changed the way developers manage code.

A2: License Analysis — GPL v2

Git is licensed under the GNU General Public License version 2 (GPL v2). This license allows users to use, modify, and share the software freely. One important feature is “copyleft,” which means any modified version must also remain open-source. This keeps the software free for everyone. It also prevents companies from making it private. GPL v2 encourages sharing and collaboration among developers. It ensures transparency because source code must be available. Many big projects like Linux also use this license. Overall, GPL v2 protects user rights and promotes open-source development.

A3: Ethical Reflection

Git supports ethical practices in software development. It promotes transparency by keeping a complete history of changes. This helps in tracking who made what changes. It also encourages collaboration, where anyone can contribute equally. Open-source tools like Git provide equal opportunities to developers worldwide. There is no restriction based on company or location. It builds trust among users because everything is open and visible. Git also

supports honesty and accountability in coding. Overall, it follows the values of fairness, openness, and teamwork. These principles are very important in modern technology.

Part B — Linux Footprint

WSL Installation and Setup

```
dixita_nema@LAPTOP-L9HT61AT:~$ uname
Linux
```

```
dixita_nema@LAPTOP-L9HT61AT:~$ git --version
git version 2.43.0
```

```
dixita_nema@LAPTOP-L9HT61AT:~$ sudo apt update
sudo apt install git
[sudo] password for dixita_nema:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7348 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1060 kB in 5s (210 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

We installed the Windows Subsystem for Linux. We set up a Linux environment.This environment is used to run Bash scripts.

Part C — FOSS Ecosystem

Git is an important part of the Free and Open Source Software (FOSS) ecosystem. It is used by platforms like GitHub and GitLab for managing code. Git helps developers collaborate and share their work easily.

It is written in the C programming language and uses system libraries. The Git community includes developers from all over the world. They contribute by improving features and fixing issues. Git plays a key role in version

control and project management. It supports open-source development and teamwork. Without Git, managing large projects would be difficult. It is a backbone of modern software development.

Part D — Open Source vs Proprietary

Here is a comparison of source and proprietary software:

Dimension	Open Source	Proprietary Alternative
Cost	Free	Paid
Security (who can audit the code?)	Transparent	Closed
Support and reliability	Community	Company
Freedom to modify	High	Limited
Community vs corporate control	User-Controlled	Vendor-Controlled
Your overall verdict		

Linux Environment and Bash Scripting

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Script Implementation

Script 1: System Identity Report

This script displays information about the Linux system, including the current user, hostname, operating system details, kernel version, system uptime and hardware information such as CPU and memory. It helps in understanding the Linux system configuration and environment.

Code:

```
#!/bin/bash

# Script 1: Advanced System Identity Report


clear


echo "==============================================="

echo "       OPEN SOURCE SYSTEM REPORT"

echo "==============================================="


STUDENT_NAME="Dixita nema"

SOFTWARE="Git"


KERNEL=$(uname -r)

HOSTNAME=$(dixita)

USER_NAME=$(dixita-01-nema)

HOME_DIR=$HOME

UPTIME=$(uptime -p)

DATE=$(date)

ARCH=$(uname -m)


if command -v lsb_release &>/dev/null; then

   DISTRO=$(lsb_release -d | cut -f2)

else

   DISTRO="Unknown Distribution"

fi


CPU=$(grep -m1 'model name' /proc/cpuinfo | cut -d ':' -f2)


MEM=$(free -h | awk '/Mem:/ {print $2}')
```

```
DISK=$(df -h / | awk 'NR==2 {print $2}')

echo "Student Name   : $DIXITA_NEMA"
echo "Software       : $SOFTWARE"
echo "Hostname       : $HOSTNAME"
echo "Kernel Version : $KERNEL"
echo "Architecture   : $ARCH"
echo "Distribution   : $DISTRO"
echo "User           : $dixita-01-nema"
echo "Home Directory : $HOME_DIR"
echo "System Uptime  : $UPTIME"
echo "Date & Time    : $DATE"
echo "CPU            : $CPU"
echo "Memory         : $MEM"
echo "Disk Space     : $DISK"
echo "License        : GNU/Linux (GPL v2)"

echo "==============================================="
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

===============================================
        OPEN SOURCE SYSTEM REPORT
===============================================
./script1_system_identity.sh: line 17: dixita-01-nema: command not found
Student Name   :
Software       : Git
Hostname       : LAPTOP-L9HT61AT
Kernel Version : 6.6.87.2-microsoft-standard-WSL2
Architecture   : x86_64
Distribution   : Ubuntu 24.04.4 LTS
User           :
Home Directory : /home/dixita_nema
System Uptime  : up 1 hour, 1 minute
Date & Time    : Sun Mar 29 18:36:47 UTC 2026
CPU            :  AMD Ryzen 5 5500U with Radeon Graphics
Memory         : 7.4Gi
Disk Space     : 1007G
License        : GNU/Linux (GPL v2)
===============================================
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$
```

Script 2: Package Inspector

This script checks whether Git is installed on the Linux system. If installed it displays the version and additional package details using the system's package manager. It is useful for verifying software availability and system setup.

Code:

```
#!/bin/bash

# Script 2: Advanced Package Inspector (Git)


PACKAGE="git"


echo "======================================="

echo "   FOSS PACKAGE INSPECTOR"

echo "======================================="


if command -v $PACKAGE &>/dev/null; then

   echo "[✓] $PACKAGE is installed"

   VERSION=$($PACKAGE --version)

   echo "Version: $VERSION"


   echo "---------------------------------------"


   if command -v dpkg &>/dev/null; then

      echo "Using dpkg (Debian-based system)"

      dpkg -s $PACKAGE | grep -E 'Version|Maintainer'

   elif command -v rpm &>/dev/null; then

      echo "Using rpm (RedHat-based system)"

      rpm -qi $PACKAGE | grep -E 'Version|License|Summary'

   else

      echo "Unknown package manager"

   fi

else

   echo "[✗] $PACKAGE is NOT installed"

fi


echo "---------------------------------------"

echo "Open Source Philosophy Notes:"
```

```
case $PACKAGE in
   git) echo "Git: Decentralized collaboration tool enabling freedom in development";;
   apache) echo "Apache: Backbone of the open web";;
   mysql) echo "MySQL: Open-source database powering applications";;
   vlc) echo "VLC: Media freedom for everyone";;
   *) echo "General FOSS tool";;
esac

echo "======================================="
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

=============================================
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$ ./script2_package_inspector.sh
=====================================
   FOSS PACKAGE INSPECTOR
=====================================
[✓] git is installed
Version: git version 2.43.0
-------------------------------------
Using dpkg (Debian-based system)
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Version: 1:2.43.0-1ubuntu7.3
Original-Maintainer: Jonathan Nieder <jrnieder@gmail.com>
-------------------------------------
Open Source Philosophy Notes:
Git: Decentralized collaboration tool enabling freedom in development
=====================================
```

Script 3: Disk and Permission Auditor

This script analyzes directories in the Linux system and displays their permissions, ownership and disk usage. It helps in monitoring storage usage and identifying permission-related issues.

Code:

```
#!/bin/bash

# Script 3: Advanced Disk & Permission Auditor


DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")


echo "======================================="

echo "   DISK & PERMISSION AUDIT REPORT"

echo "======================================="
```

```
for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1}')
        OWNER=$(ls -ld $DIR | awk '{print $3}')
        GROUP=$(ls -ld $DIR | awk '{print $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

        echo "Directory: $DIR"
        echo "Permissions: $PERMS"
        echo "Owner: $OWNER"
        echo "Group: $GROUP"
        echo "Size: $SIZE"
        echo "--------------------------------------"
    else
        echo "$DIR not found"
    fi
done

# Git Binary Check
echo "Checking Git Binary Location..."
which git

echo "======================================="
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$ ./script3_disk_permission_auditor.sh
=====================================
   DISK & PERMISSION AUDIT REPORT
=====================================
Directory: /etc
Permissions: drwxr-xr-x
Owner: root
Group: root
Size: 4.3M
-------------------------------------
Directory: /var/log
Permissions: drwxr-xr-x
Owner: root
Group: syslog
Size: 17M
-------------------------------------
Directory: /home
Permissions: drwxr-xr-x
Owner: root
Group: root
Size: 28K
-------------------------------------
Directory: /usr/bin
Permissions: drwxr-xr-x
Owner: root
Group: root
Size: 95M
-------------------------------------
```

```
-------------------------------------
Directory: /tmp
Permissions: drwxrwxrwt
Owner: root
Group: root
Size: 28K
-------------------------------------
Checking Git Binary Location...
/usr/bin/git
=====================================
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$
```

Script 4: Log File Analyzer

This script reads a specified log file and searches for a given keyword (default is "error”). It counts the number of occurrences. Displays the most recent matching lines helping in debugging and system monitoring.

Code:

```
#!/bin/bash

# Script 4: Advanced Log Analyzer


LOGFILE=$1

KEYWORD=${2:-error}

COUNT=0


if [ -z "$LOGFILE" ]; then

   echo "Usage: $0 <logfile> [keyword]"

   exit 1
```

```
fi

if [ ! -f "$LOGFILE" ]; then
    echo "File not found. Trying alternative logs..."

    for ALT in /var/log/syslog /var/log/messages; do
        if [ -f "$ALT" ]; then
            LOGFILE=$ALT
            echo "Using $LOGFILE"
            break
        fi
    done
fi

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT+1))
    fi
done < "$LOGFILE"

echo "-------------------------------------"
echo "Keyword '$KEYWORD' found $COUNT times"
echo "-------------------------------------"

echo "Last 5 matches:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```

Output:

```
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$ ./script4_log_analyzer.sh /var/log/sy
slog
--------------------------------------
Keyword 'error' found 8 times
--------------------------------------
Last 5 matches:
2026-03-29T17:35:32.225654+00:00 LAPTOP-L9HT61AT kernel: WSL (248) ERROR: CheckConnection: getaddrinfo() failed: -5
2026-03-29T17:35:32.484599+00:00 LAPTOP-L9HT61AT snapd[203]: helpers.go:190: error trying to compare the snap system key: system-ke
y missing on disk
2026-03-29T17:35:33.992929+00:00 LAPTOP-L9HT61AT wsl-pro-service[207]: #033[33mWARNING#033[0m Could not ensure valid Landscape conf
iguration: could not ensure valid Landscape configuration: could not register distro to Landscape: could not enable Landscape: /usr
/bin/landscape-config: error: exit status 1.
2026-03-29T17:35:40.496922+00:00 LAPTOP-L9HT61AT kernel: WSL (2 - init-systemd(Ubuntu)) ERROR: WaitForBootProcess:3432: /sbin/init
failed to start within 10000ms
2026-03-29T17:46:57.533068+00:00 LAPTOP-L9HT61AT wsl-pro-service[207]: #033[33mWARNING#033[0m Exiting after could not ensure valid
Landscape configuration: could not register distro to Landscape: could not enable Landscape: /usr/bin/landscape-config: error: exit
 status 1.
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$
```

Script 5: Manifesto Generator

This script is like a conversation with the user. It asks for some information. Then creates a special open-source manifesto. The result is saved to a text file. Shown on the screen. This shows how to handle user input and work with files in Bash.

Code:

```
#!/bin/bash

# Script 5: Open Source Manifesto Generator


echo "======================================="

echo " OPEN SOURCE MANIFESTO GENERATOR"

echo "======================================="


while [ -z "$TOOL" ]; do

   read -p "1. Name an open-source tool: " TOOL

done


while [ -z "$FREEDOM" ]; do

   read -p "2. What does freedom mean to you? " FREEDOM

done


while [ -z "$BUILD" ]; do

   read -p "3. What will you build and share? " BUILD

done
```

```
DATE=$(date '+%d %B %Y')

USER=$(whoami)

OUTPUT="manifesto_$USER.txt"


echo "--------------------------------------"


echo "On $DATE, I believe in open source." > $OUTPUT

echo "I regularly use $TOOL." >> $OUTPUT

echo "For me, freedom means $FREEDOM." >> $OUTPUT

echo "I aim to build $BUILD and share it with the world." >> $OUTPUT


echo "--------------------------------------"

echo "Manifesto saved to $OUTPUT"

echo "--------------------------------------"


cat $OUTPUT
```

Output:

```
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$ ./script5_manifesto_generator.sh
======================================
 OPEN SOURCE MANIFESTO GENERATOR
======================================
1. Name an open-source tool: Linux
2. What does freedom mean to you? Exploring
3. What will you build and share? Strength and Kindness
--------------------------------------
--------------------------------------
Manifesto saved to manifesto_dixita_nema.txt
--------------------------------------
On 29 March 2026, I believe in open source.
I regularly use Linux.
For me, freedom means Exploring.
I aim to build Strength and Kindness and share it with the world.
dixita_nema@LAPTOP-L9HT61AT:/mnt/c/Users/Dixita Nema/OneDrive/Documents/OSS-Audit-24BAI10802$
```

Learning Outcomes

From this project, I learned many useful things about open-source software and Linux. I understood how to use Linux commands and manage files. I also learned how to write Bash scripts to automate tasks. Knowledge about file permissions and system structure improved my understanding. I learned about open-source licenses like GPL v2. Git helped me understand version control and teamwork. These skills are useful for real-world software development. The project improved my problem-solving ability. It also gave me practical experience along with theory. Overall, it was a great learning experience.

Challenges Faced

While working on this project, I faced some challenges. Setting up WSL was difficult at first due to installation issues. Understanding Linux file paths was also confusing. I made mistakes while writing Bash scripts, which caused errors. Debugging those errors took time and effort. Managing file permissions using commands like chmod was tricky. But with practice, I improved my skills. These challenges helped me learn better. They also made me more confident in handling problems. Overall, the difficulties turned into learning opportunities.

Conclusion

This project helped me understand Git and open-source software in a practical way. I learned how Git is used in real-life projects. Working with Linux and Bash scripts gave me hands-on experience. I understood the importance of collaboration in development. The project connected my theory knowledge with practical skills. It also improved my technical understanding. Git plays a very important role in software development. Overall, this project was very helpful and informative. It prepared me for future learning and real-world applications.